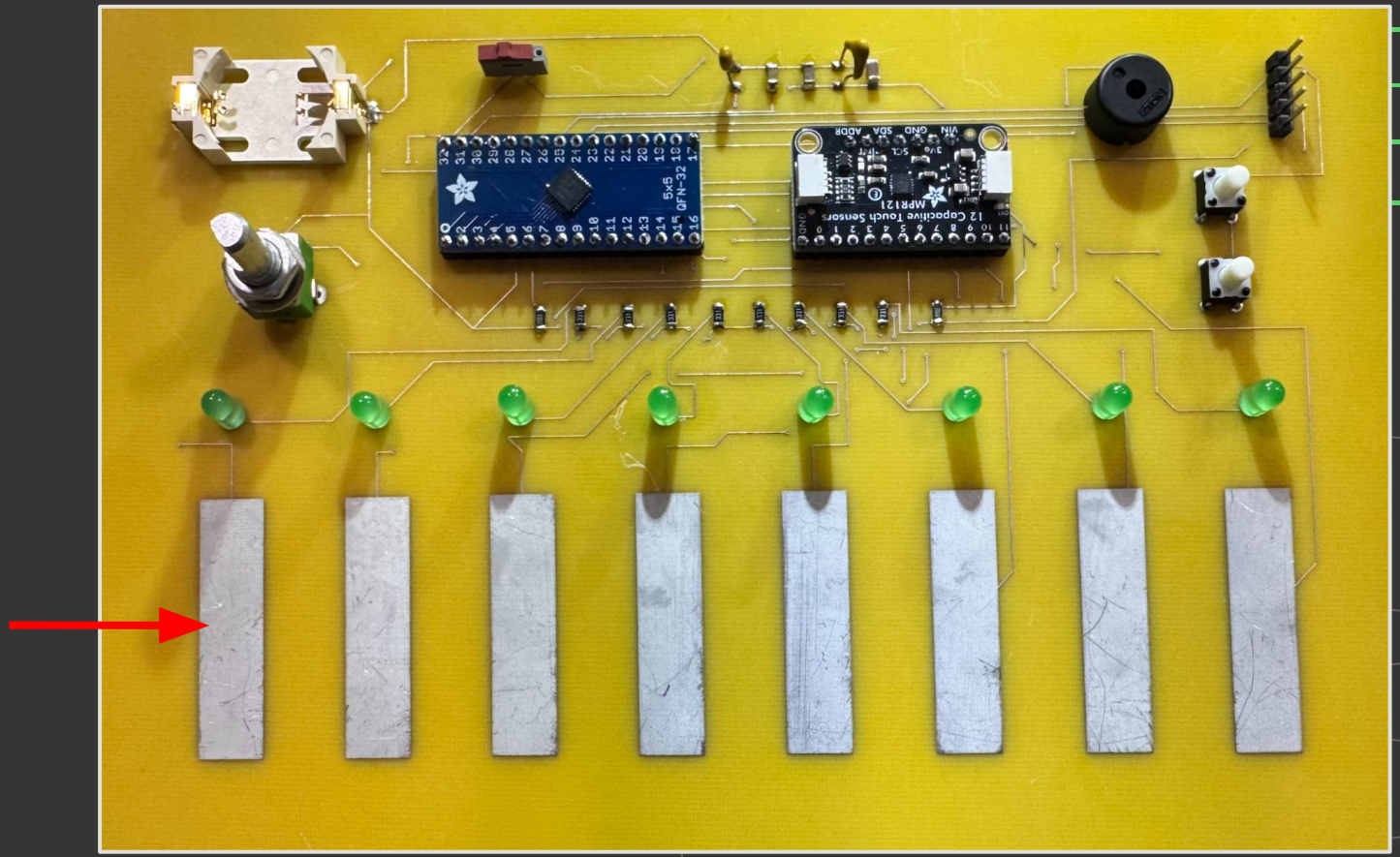
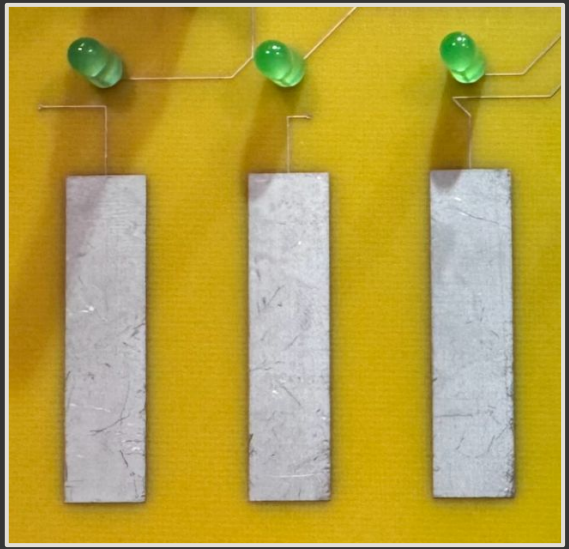
Eugenia Sung & Anjali Sivasothy
ELEC 327 Final Project

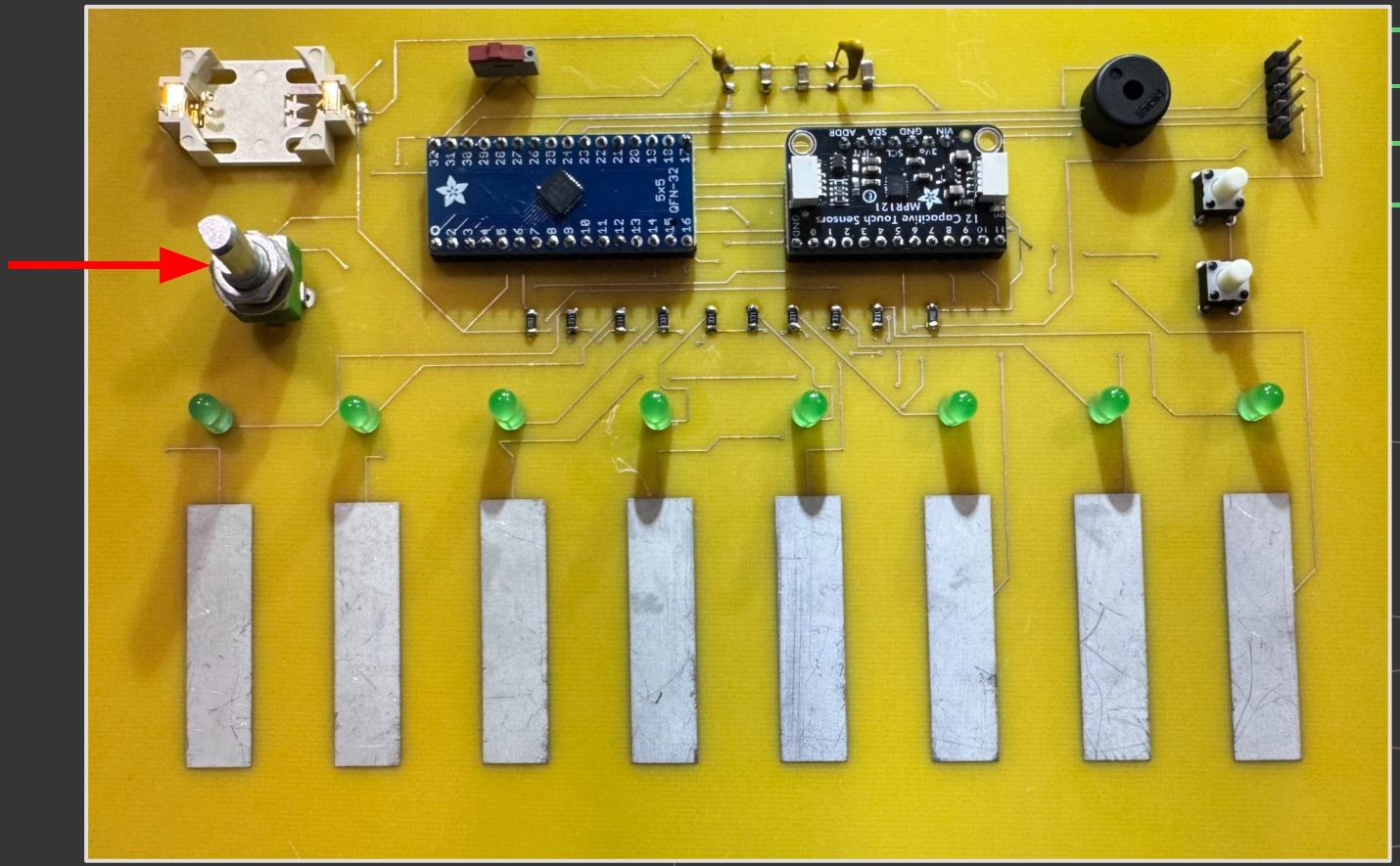
# Capacitive Touch Piano

Touch metal. Play music.

# The Features


5x5
QFN-32
MPR121
12 Capacitive Touch Sensors
VIN GND SDA ADDR
3/6 SCL IRQ

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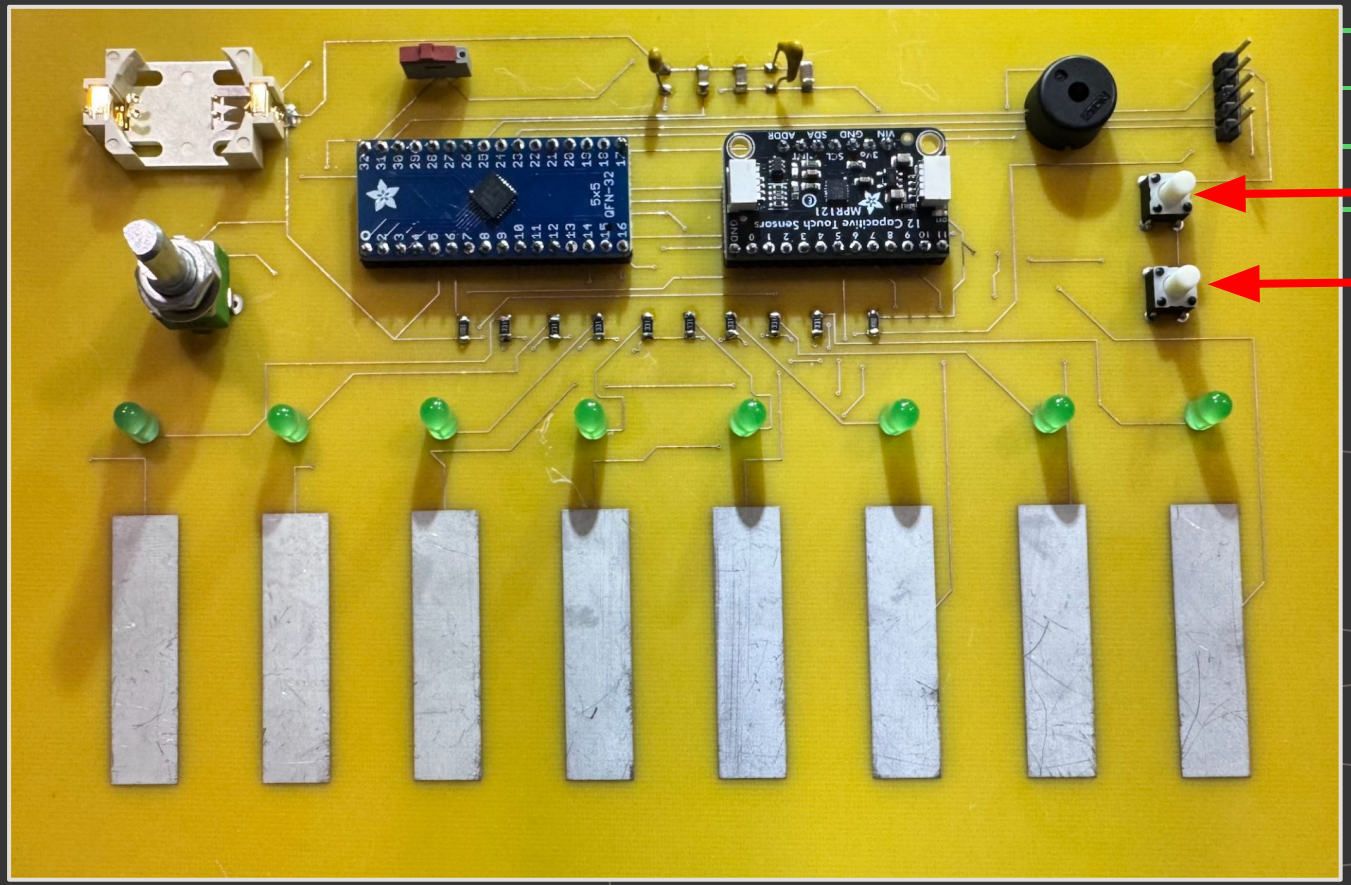
DISP
VOL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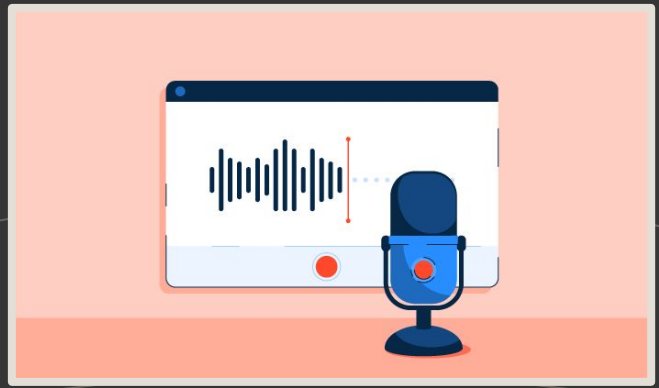
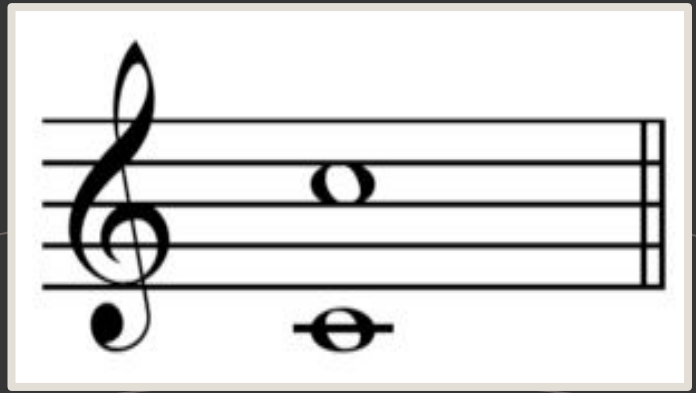
GND





# How Does it *Work*?

# How Do the Keys Work?

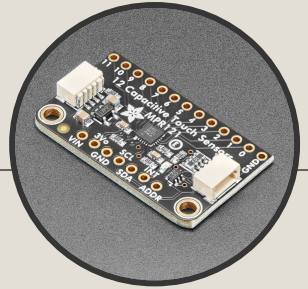
**Step 1**

User Finger

**Step 2**

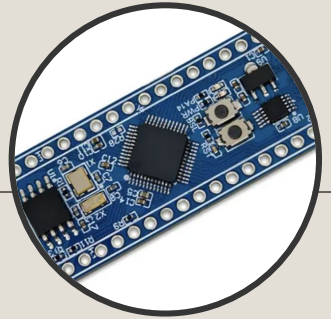
Copper Pad

**Step 3**



MPR121 Sensor

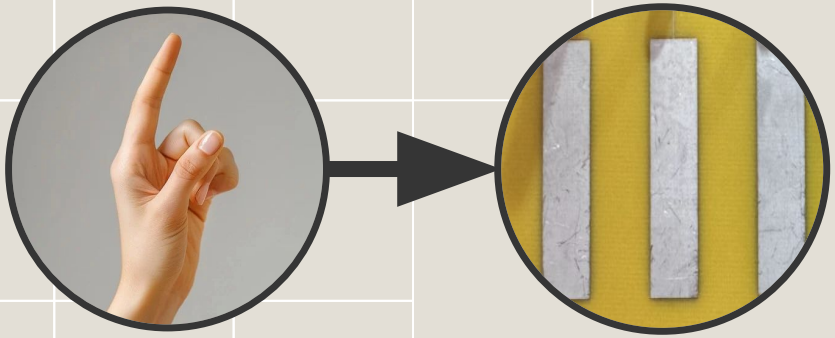
**Step 4**



MSPM0 Microcontroller

**Step 5**

Buzzer and LED Output

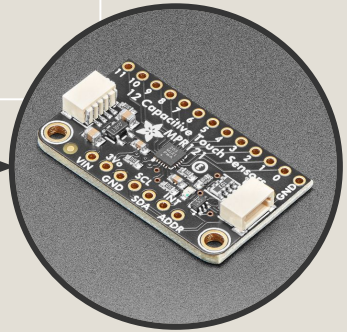


User Finger

Copper Pad

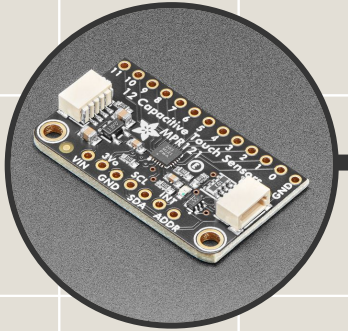
The copper pad, when powered, creates an electric field. When your finger enters that field, it **increases the capacitance of the pad.**

Copper Pad

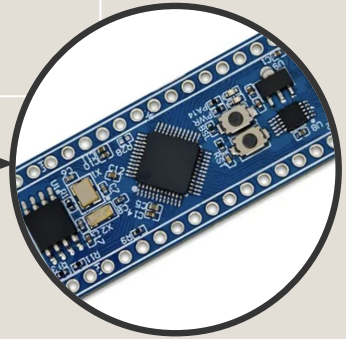



MPR121 Sensor

The MPR121 continuously monitors the capacitance of each copper pad. When capacitance **increases** above a set threshold, **a touch is registered.**

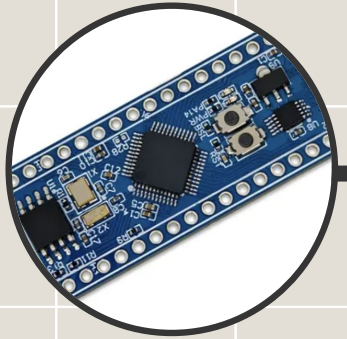



**MPR121 Sensor**



**MSPM0 Microcontroller**

Once MPR121 detects a touch, the MSPM0 reads the sensor over I²C. **The touch data is received as a set of bits.** The active bit corresponds to the pressed key.

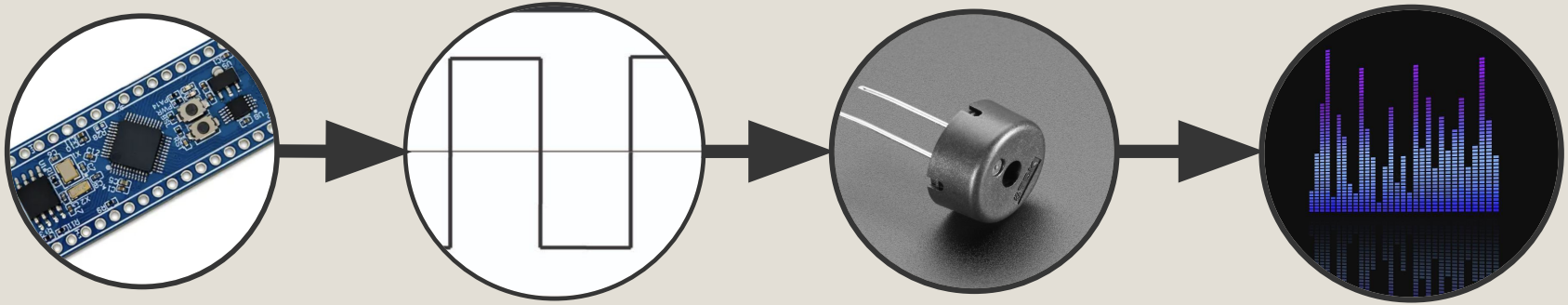


**MSPM0 Microcontroller**

**Buzzer & LED Output**

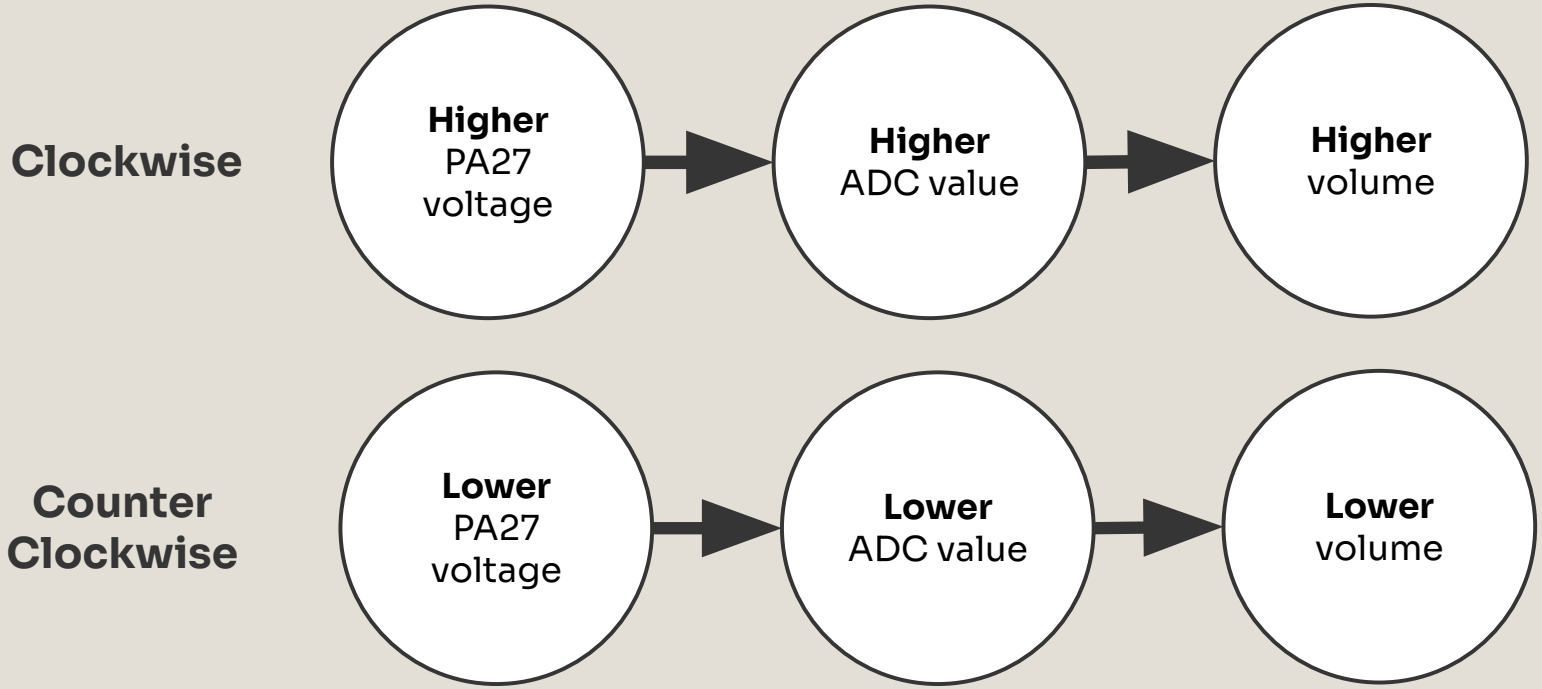
Once the MSPM0 identifies the active key, it uses **GPIO** to turn on the key's corresponding **LED** and drive the **buzzer** at the key's assigned frequency.

# How Does the Buzzer Make Sound?

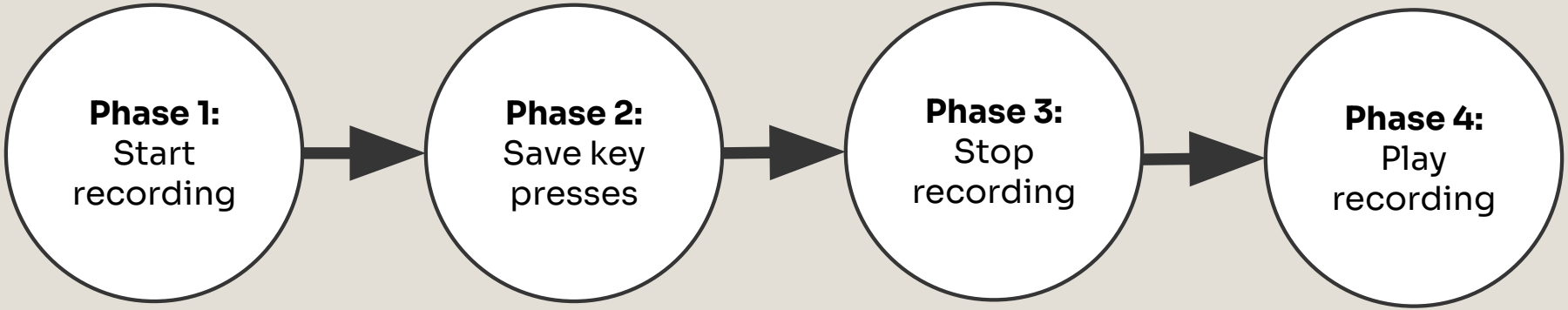


**MSPM0 Microcontroller** → **Square Wave** → **Buzzer** → **Sound**

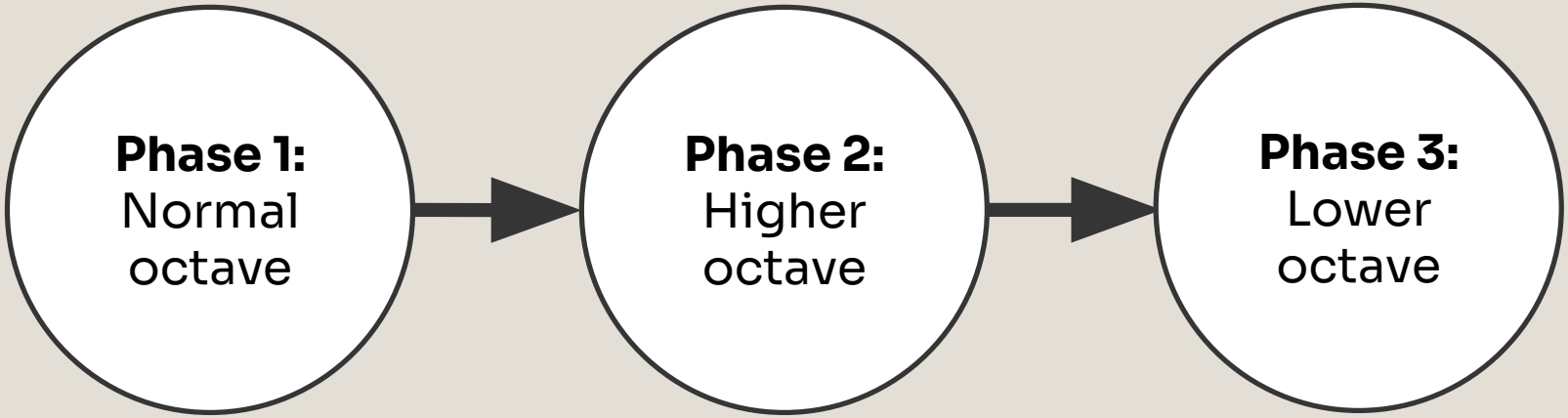
# How Does the Potentiometer Work?



# How Does the Playback Button Work?



# How does Octave Button Work?

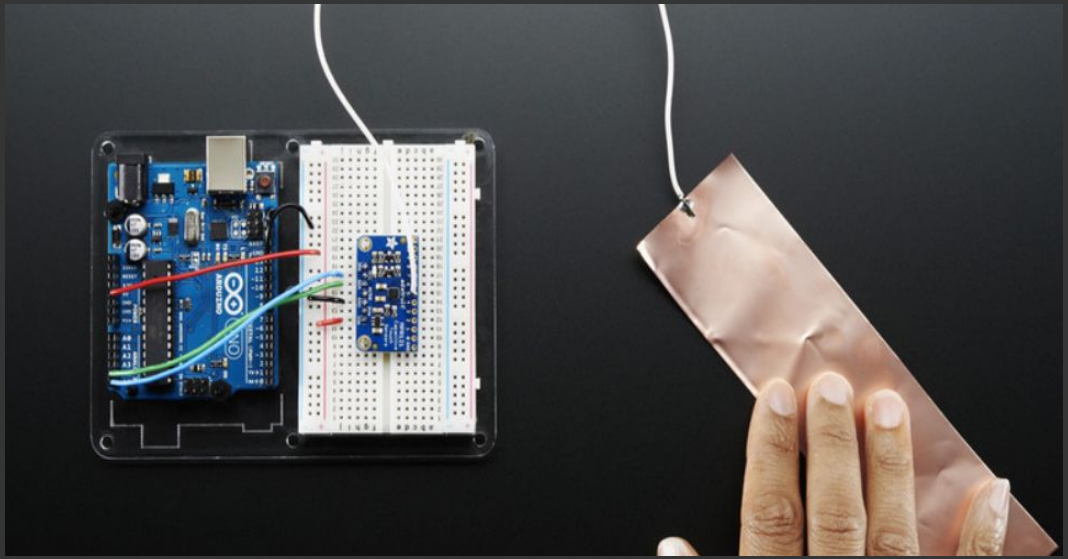


# Design Decisions

# Custom Capacitive Sensing is Hard

- Optimal electrode geometry?
- Extremely sensitive to noise and layout
- What counts as noise vs a real touch?
- Will this even work after fabrication?

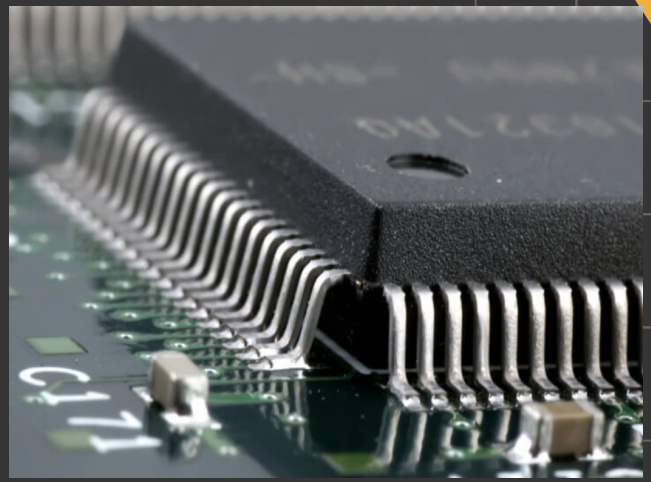
# Using the MPR121



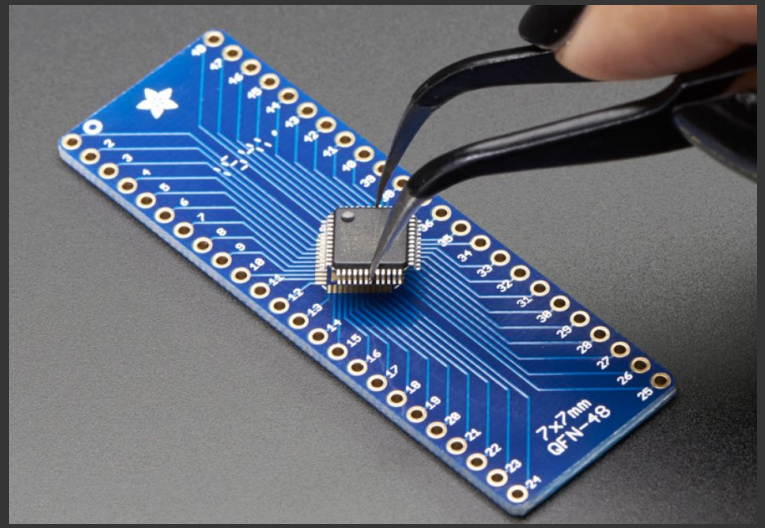
Reliably offloads capacitive sensing

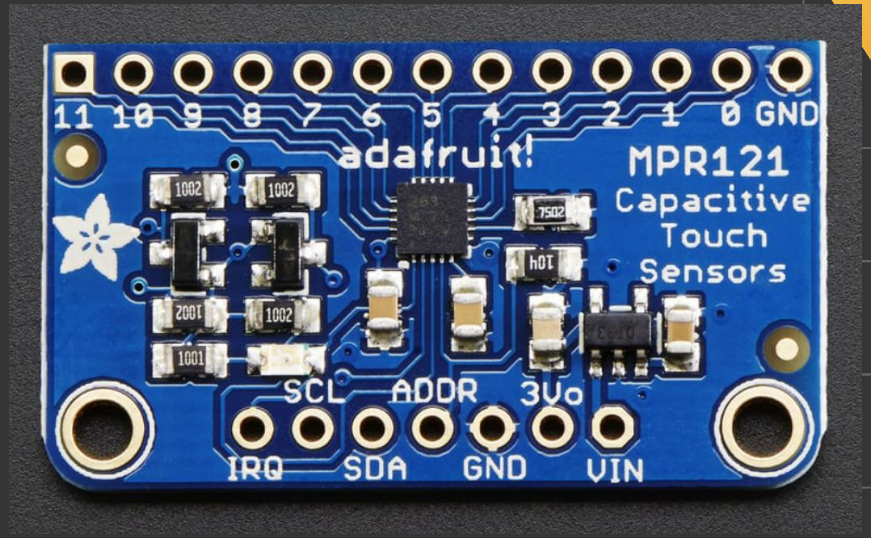
# Soldering is Hard

- Fine-pitch ICs are difficult to solder by hand
- High risk of bridges/poor connections
- A reliable first pass design?



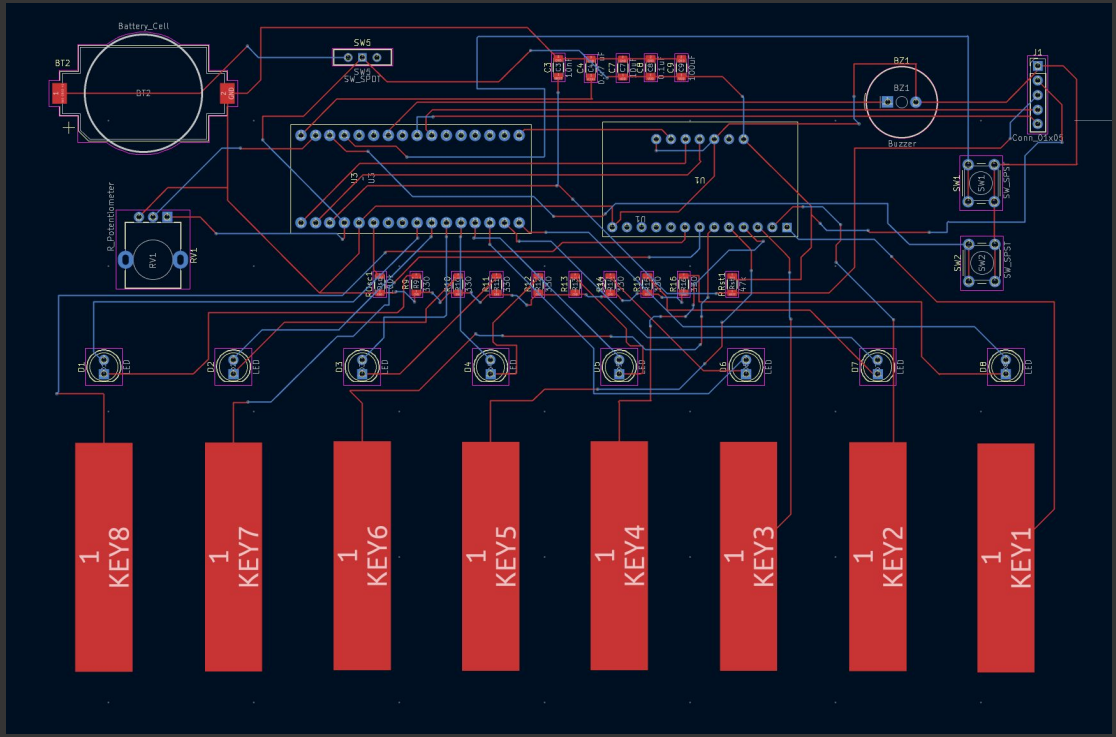
# Breakout Boards

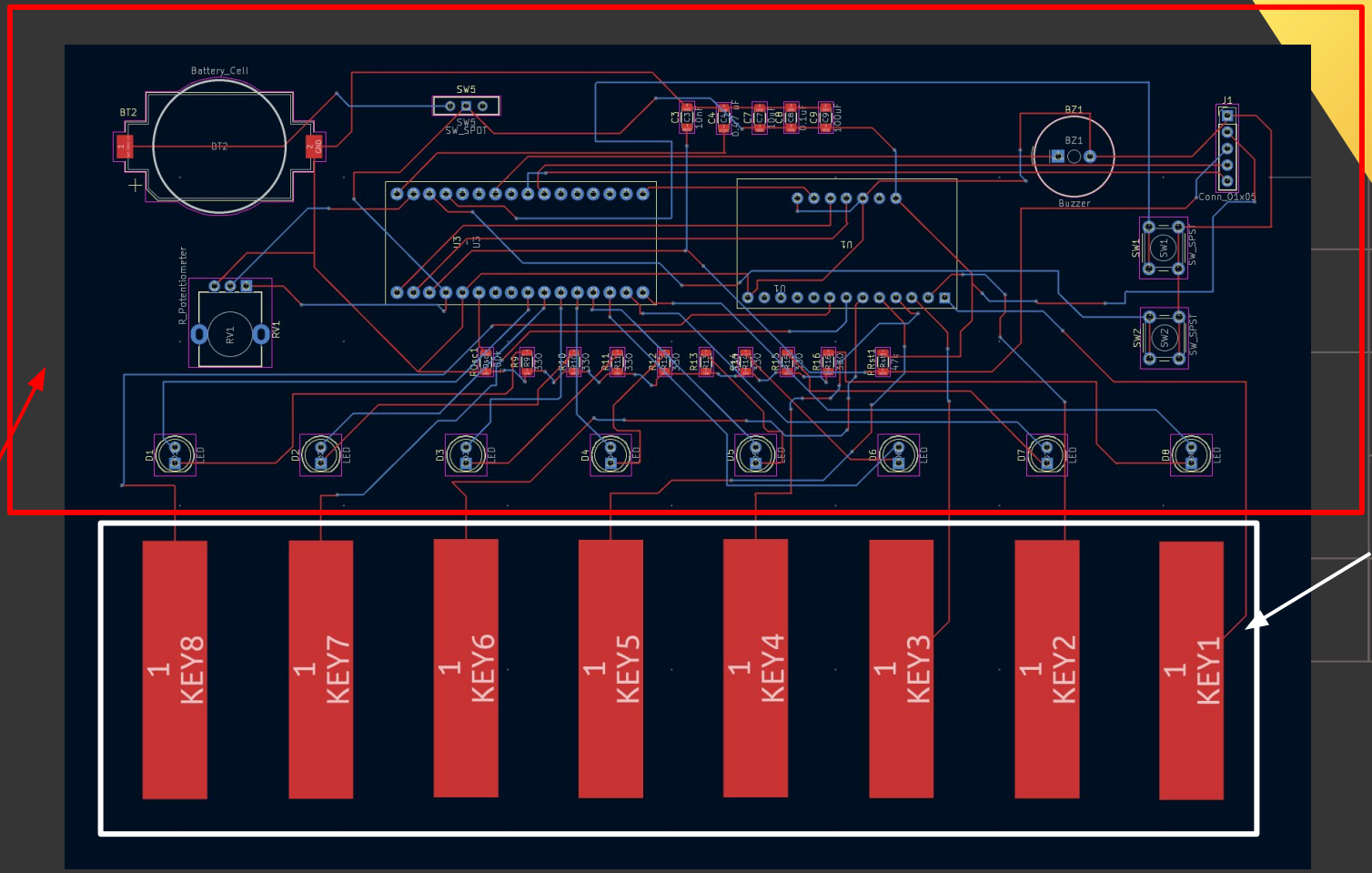



Easier assembly and built in support

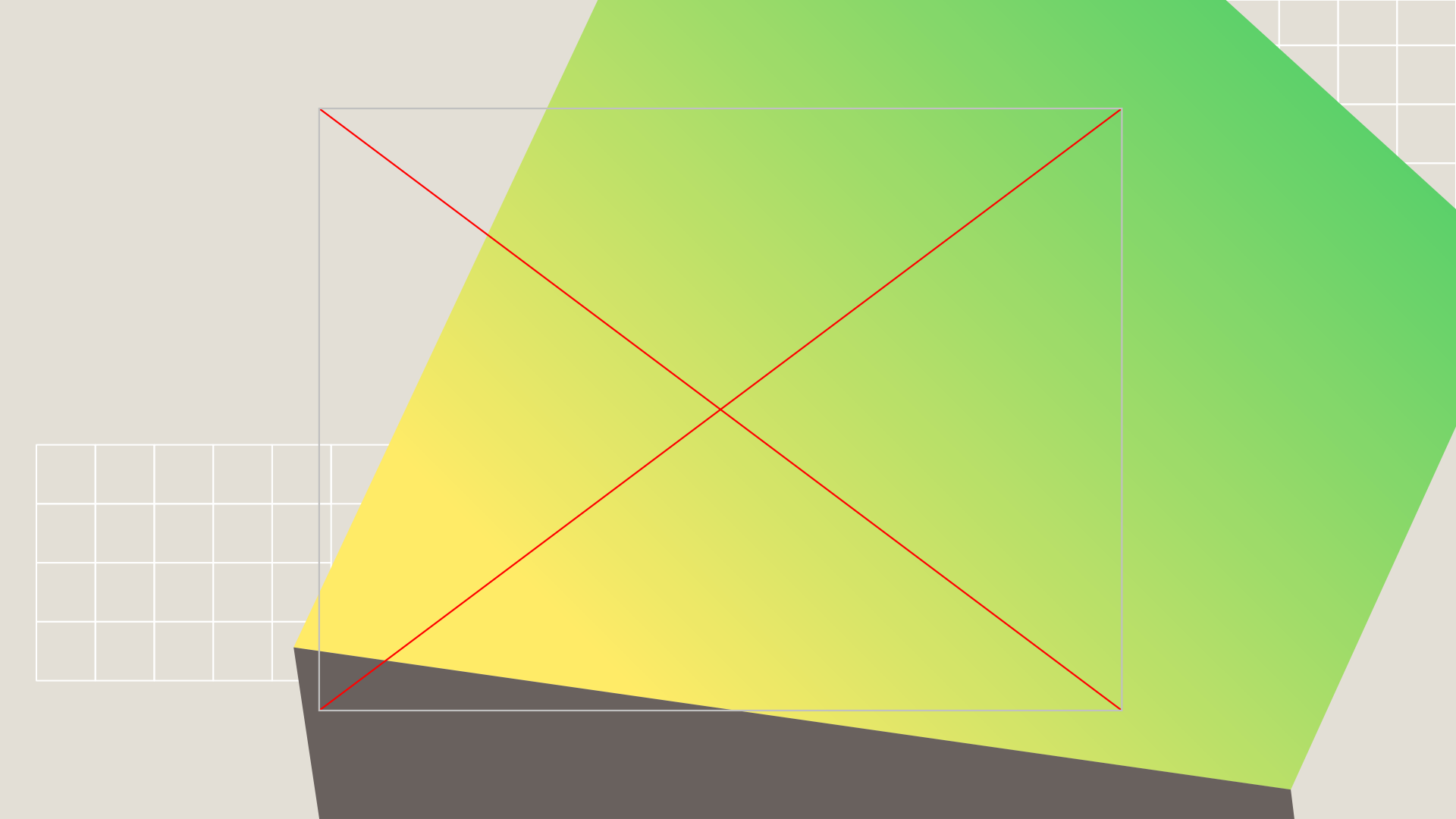
# Mixed Signal PCB Layout






High Noise Region

Low Noise Region

# Future Work & Demo

- Integrate MSPM0 and MPR121 directly (no breakouts)
- Improve audio output
- Support multiple key presses at once
- Power optimization